

Module A - Technical Selection Criteria

Smart Digital Stethoscope | KAAAN-A1 / KAAAN-A2

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In this note, I evaluated microphone selection criteria, sources of interference, and phantom requirements for heart sound acquisition. In line with the project plan, we plan to compare one electret model and one MEMS model. Including spares, **2 units of each model** are required; suitable laboratory stock will be deducted from this requirement. At this stage, the work is based on document review; physical tests have not yet been performed. Experiments will be conducted only on the phantom. [1]

1 Frequency range and low-frequency requirement

The project characterization band is **20-500 Hz**. The component requirements document particularly emphasizes the region around **20-150 Hz**. I therefore prioritized low-frequency response as a selection criterion to preserve the S1/S2 structure. The analysis band will not be limited to 150 Hz. The 20-220 Hz bell mode in the T1 example also supports the low-frequency requirement. [1,3]

A specified range of 20 Hz-20 kHz does not indicate a flat response at 20 Hz or a lower -3 dB point of 20 Hz. The lower -3 dB point is where the response decreases by 3 dB relative to the stated reference. I prefer a documented lower cutoff at or below 20 Hz, preferably at or below 10 Hz; this is my selection recommendation, not a mandatory acceptance threshold in the project plan. A first-order 100 Hz high-pass filter attenuates 20 Hz by approximately 14 dB. Therefore, filters in the analog front end (AFE), ADC/codec, and playback path will also be examined along with the microphone. [2; calculation]

2 Microphone datasheet evaluation criteria

Item	Properties I will evaluate
Low-frequency response	Response at 20, 30, 50, 100, 150, and 500 Hz; graph, tolerance limits, and measurement conditions. The typical curve and guaranteed limits will be evaluated separately.
Lower -3 dB point	The value and source page will be recorded; if unspecified, this will be noted. Estimates from graphs will be marked separately. Internal high-pass/DC removal filters will be examined.
Sensitivity and limits	Reference frequency and sound pressure, tolerance, unit, THD/AOP, and clipping headroom. Analog mV/Pa or dBV/Pa values will not be compared directly with digital dBFS.
Noise and self-noise	SNR/EIN, weighting, and measurement band. In addition to catalog SNR, unweighted system noise will be measured at 20-500 Hz, particularly at 20-100 Hz.
Supply and interface	Electret: bias, load resistance, output impedance, AFE/ADC. MEMS: analog/PDM/I ² S output, voltage, clock, data alignment, actual resolution, current, and supply noise.
Package and mounting	Acoustic port orientation, dimensions, connection/solder layout, and carrier board. Keeping the port open; checking the seal, acoustic cavity, leakage, and cable fastening.

General definitions [2,4]; project measurement scope [1]. High sensitivity or digital output alone does not imply a better heart sound recording. MEMS models can also have analog outputs.

3 Sources of interference and planned controls

Contact/motion: Friction and cable pulling can cause transients, baseline shifts, and clipping. I plan to use a fixed holder, cable fastening, and a controlled motion experiment. **Speech/environment:** A quiet recording and an ambient sound experiment at a fixed distance/level will be performed for noise that can overlap the heart sound band. **Mains:** Short shielded analog connections, an appropriate return path, and supply conditions will be evaluated for 50 Hz and its harmonics; the notch filter will remain disabled during raw measurements. **Mechanical coupling:** Pressure, position, seal, acoustic cavity, and silicone thickness will be recorded, and the variables will be examined individually. [1,5]

4 Selection process

Week 3: The exact part/board numbers, quantities, and connection requirements of the two models to be compared will be determined. **Weeks 10-11:** The acquisition path for the final prototype will be selected through phantom measurements. The current models are preliminary candidates; their final suitability has not yet been verified. [1]

Phantom Setup and Candidate Evaluation

Module A | Experiment preparation and proposals for the Week 3 meeting

5 Initial phantom setup

Component	Technical requirement and planned check
Exciter / small transducer	Stable excitation without clipping at 20-500 Hz; verification of sufficient output, particularly at 20-50 Hz. Load impedance and drive limits will be evaluated in addition to dimensions and power specifications.
Amplifier and playback	Power/voltage suitable for the exciter, fixed gain, and an input that preserves low frequencies. Automatic gain, EQ, and noise suppression will remain disabled; PC output, capacitors, supply, and distortion will be checked.
Silicone / gel layer	Material, thickness, hardness, contact area, and compression will be recorded; gaps and slipping will be prevented. I consider the 3 mm thickness in REV01 an initial candidate.
Rigid frame and chestpiece	Position marks, access for sensor replacement, repeatable pressure, and a cable holder will be used. A fixed setup will be established for comparison instead of pressing the chestpiece by hand.
Reference and recording	The source WAV, output level, physical setup, acquisition settings, and raw recordings will be retained. Without an independent reference, the result will be interpreted as the combined response of the phantom and acquisition chain, rather than the microphone alone.

The component groups are based on project requirements [1]. CirCor recordings will be played at the correct rate; if conversion from 4 kHz to 48 kHz is needed, resampling will be performed instead of changing the file header. [8]

6 Microphone candidates and information to be completed

Preliminary candidate / total need	My current assessment and next step
Electret: Kingstate KECG2740PBJ 2 units	The response curve starts at approximately 50 Hz. I will seek additional manufacturer information on the 20-50 Hz response and lower -3 dB point. I will clarify the AFE and ADC/reference audio interface path with Ozan. [6]
MEMS: PUI DMM-4026-B-I2S-R 2 units; EB-R board in the BOM	The typical curve extends to 20 Hz; no explicit lower -3 dB value is given. I will examine low-frequency information, while Ozan will examine clock/data format and board compatibility. The DMM-4026-B-I2S-EB-R board will be the basis for procurement. [7]

I consider both candidates conditional. Missing low-frequency information will be sought through manufacturer documentation or clarification. If it cannot be obtained, the uncertainty will be evaluated at the meeting, or a better-documented candidate will be selected. With the available data, I cannot conclude that performance at 20 Hz has been verified.

7 Meeting preparation and final decision method

Preparation: The team will check the models, quantities, and availability of laboratory components. I am responsible for preparing the technical information gaps for the microphones and the phantom mounting proposal. Ozan will prepare board/connection information, stock availability, and prices; Ege will prepare raw recording, interface, and test requirements. **Meeting objective:** Determine the two microphone models and quantities, MCU, phantom components, initial data packet draft, and BOM v1.0 purchase list. The team plan aims to complete ordering before Week 4. [1,9]

My proposed comparison: Under the same phantom, excitation, and pressure conditions, with the gain of each path documented and held constant, the 20-500 Hz response, defined SNR, clipping, and spread across at least 10 repetitions will be measured. Acceptance thresholds will be set before measurement. Suitable candidates will be compared in terms of low-frequency response, noise, and stability. If the difference remains within measurement uncertainty, additional measurements will be performed, or the selection will be justified in terms of integration and cost. [1]

References

[1] SDP4 Digital Stethoscope Plan, Sec. 4, 7-8; SDP4 Component Requirements, Group A and Notes Before Ordering; REV01 System Design. [2] Analog Devices, [AN-1112: Microphone Specifications Explained](#). [3] Texas Instruments, [SPRAB38A](#). [4] TDK InvenSense, [Low Self Noise](#). [5] Zauli et al., [Exploring Microphone Technologies for Digital Auscultation Devices](#), 2023. [6] Kingstate KECG2740PBJ, pp. 2-3. [7] PUI DMM-4026-B-I2S-R, Rev A, pp. 1, 3. [8] PhysioNet, [CirCor v1.0.3](#). [9] Team task plan: TAKIM-01, TAKIM-02, TAKIM-03. Technical source review date: 01.10.2026.